PROJECTIVE SPACE IS A PROJECTIVE BUNDLE

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Relative Proj commutes with base change. lemma-relative-proj-base-change (same chapter, judged non-self-contained this batch, not carded). Pulls the homogeneous spectrum back along g: S to Spec(Z).
- Symmetric algebra identification. definitional identity. O_S[T_0..T_n] with the standard grading is Sym of O_S^{n+1}.

Source reference. Original lemma 01O3: sources/stacks-constructions/pinned-original-source.tex, source lines 4298–4298. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $n \geq 0$. Then $\mathbf{P}^n_S$ is a projective bundle over S.*

Proof. Note that

$$\mathbf{P}^n_{\mathbf{Z}} = \mathrm{Proj}(\mathbf{Z}[T_0, \ldots, T_n]) = \underline{\mathrm{Proj}}_{\mathrm{Spec}(\mathbf{Z})}\left(\widetilde{\mathbf{Z}[T_0, \ldots, T_n]}\right)$$

where the grading on the ring $\mathbf{Z}[T_0, \ldots, T_n]$ is given by $\deg(T_i) = 1$ and the elements of $\mathbf{Z}$ are in degree 0. Recall that $\mathbf{P}^n_S$ is defined as $\mathbf{P}^n_{\mathbf{Z}} \times_{\mathrm{Spec}(\mathbf{Z})} S$. Moreover, forming the relative homogeneous spectrum commutes with base change, see Lemma 01O3. For any scheme $g : S \to \mathrm{Spec}(\mathbf{Z})$ we have $g^*\mathcal{O}_{\mathrm{Spec}(\mathbf{Z})}[T_0, \ldots, T_n] = \mathcal{O}_S[T_0, \ldots, T_n]$. Combining the above we see that

$$\mathbf{P}^n_S = \underline{\mathrm{Proj}}_S(\mathcal{O}_S[T_0, \ldots, T_n]).$$

Finally, note that $\mathcal{O}_S[T_0, \ldots, T_n] = \mathrm{Sym}(\mathcal{O}_S^{\oplus n+1})$. Hence we see that $\mathbf{P}^n_S$ is a projective bundle over S. □

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.